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THICK LENSES

When the thickness of a lens cannot be considered small compared with its focal length, some of the thin-lens formulas of Chap. 4 are no longer applicable. The lens must be treated as a *thick lens*. This term is used not only for a single homogeneous lens with two spherical surfaces separated by an appreciable distance but also for any system of coaxial surfaces which is treated as a unit. The thick lens may therefore include several component lenses, which may or may not be in contact. We have already investigated one case which comes under this category, namely, the combination of a pair of thin lenses spaced some distance apart, as shown in Fig. 4H.

5.1 TWO SPHERICAL SURFACES

A simple form of thick lens comprises two spherical surfaces as shown in Fig. 5A.

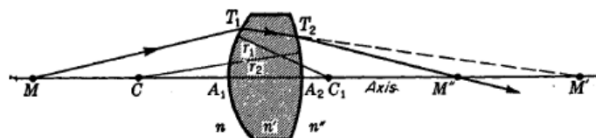


FIGURE 5A
Details of the refraction of a ray at both surfaces of a lens.

We shall first consider the parallel-ray method for graphically locating an image formed by a thick lens and then apply the general formulas already given for calculating image distances. The formulas to be used are (see Sec. 3.4)

$$\bullet \quad \frac{n}{s_1} + \frac{n'}{s_1'} = \frac{n' - n}{r_1} \quad \frac{n'}{s_2'} + \frac{n''}{s_2''} = \frac{n'' - n'}{r_2} \quad (5a)$$

For first surface For second surface

5.2 THE PARALLEL-RAY METHOD

The parallel-ray method of graphical construction, applied to a thick lens of two surfaces, is shown in Fig. 5B. Although the diagram is usually drawn as one, it has been separated into two parts here to simplify its explanation. The points F_1 and F'_1 represent the primary and secondary focal points of the first surface, and F_2 and F'_2 represent the primary and secondary focal points of the second surface, respectively.

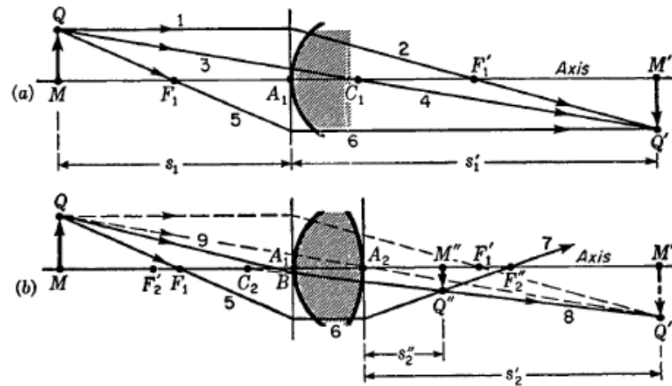


FIGURE 5B

The parallel-ray method for graphically locating the image formed by a thick lens.

Diagram (a) is constructed by applying the method of Fig. 3F to the first surface alone and extending the refracted rays as far as is necessary to locate the image $M'Q'$. This real image, $M'Q'$, then becomes the object for the second surface, as shown in diagram (b). The procedure is similar to that given for two thin lenses in Fig. 4H. Ray 5 in diagram (b), refracted parallel to the axis by the first surface, is refracted as ray 7 through the secondary focal point F'_2 of the second surface.

Rays 8 and 9 are obtained by drawing a line from Q' back through C_2 and then, through the intersection B , drawing the line BQ . The intersection of rays 7 and 8 locates the final image point Q'' and the final image $M''Q''$.

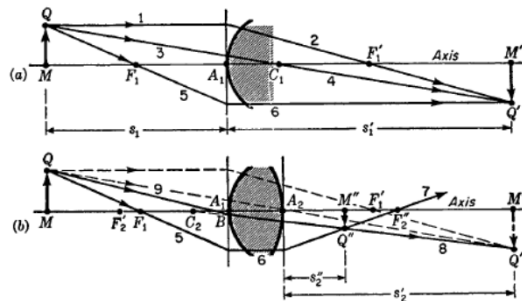


FIGURE 5B

The parallel-ray method for graphically locating the image formed by a thick lens.

EXAMPLE 1 An equiconvex lens 2 cm thick and having radii of curvature of 2 cm is mounted in the end of a water tank. An object in air is placed on the axis of the lens 5 cm from its vertex. Find the position of the final image. Assume refractive indices of 1.00, 1.50, and 1.33 for air, glass, and water, respectively.

SOLUTION The relative dimensions in this problem are approximately those shown in Fig. 5B(b). If we apply Eq. (5a) to the first surface alone, we find the image distance to be

$$\frac{1.00}{5} + \frac{1.50}{s'_1} = \frac{1.50 - 1.00}{2} \quad \text{or} \quad s'_1 = +30 \text{ cm}$$

When the same equation is applied to the second surface, we note that the object distance is s'_1 minus the lens thickness, or 28 cm, and that since it pertains to a virtual object it has a negative sign. The substitutions to be made are, therefore, $s_2 = -28 \text{ cm}$, $n' = 1.50$, $n'' = 1.33$, and $r_2 = -2.0 \text{ cm}$.

$$\frac{1.50}{-28} + \frac{1.33}{s'_2} = \frac{1.33 - 1.50}{-2} \quad \text{or} \quad s'_2 = +9.6 \text{ cm}$$

Particular attention should be paid to the signs of the various quantities in this second step. Because the second surface is concave toward the incident light, r_2 must have a negative sign. The incident rays in the glass belong to an object point M' , which is virtual, and thus s'_2 , being to the right of the vertex A_2 , must also be negative. The final image is formed in the water ($n'' = 1.33$) at a distance +9.6 cm from the second vertex. The positive sign of the resultant signifies that the image is *real*.

5.3 FOCAL POINTS AND PRINCIPAL POINTS

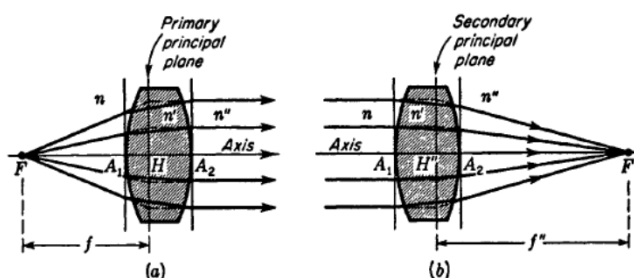


FIGURE 5C

Ray diagrams showing the primary and secondary principal planes of a thick lens.

Diagrams showing the characteristics of the two focal points of a thick lens are given in Fig. 5C. In the first diagram diverging rays from the primary focal point F emerge parallel to the axis, while in the second diagram parallel incident rays are brought to a focus at the secondary focal point F'' . In each case the incident and refracted rays have been extended to their point of intersection between the surfaces. Transverse planes through these intersections constitute *primary* and *secondary principal planes*. These planes cross the axis at points H and H'' , called the *principal points*. It will be noticed that there is a point-for-point correspondence between the two principal planes, so that each is an erect image of the other and both are the same size. For this reason they have sometimes been called *unit planes*. They are best defined by saying that *the principal planes are two planes having unit positive lateral magnification*.

The focal lengths, as shown in the figure, are measured from the focal points F and F'' to their respective principal points H and H'' and not to their respective vertices A_1 and A_2 . If the medium is the same on both sides of the lens, $n'' = n$, the primary focal length f is exactly equal to the secondary focal length f'' .

If the media on the two sides of the lens are different so that n'' is not equal to n , the two focal lengths are different and have the ratio of their corresponding refractive indices:

$$\frac{n''}{n} = \frac{f''}{f} \quad (5b)$$

In general the focal points and principal points are not symmetrically located with respect to the lens but are at different distances from the vertices. This is true even when the media on both sides are the same and the focal lengths are equal. As a lens with a given material and focal length is "bent" (see Fig. 5D), departing in either direction from the symmetrical shape of an equiconvex lens, the principal points are shifted. For meniscus lenses of considerable thickness and curvature, H and H'' may be completely outside the lens.

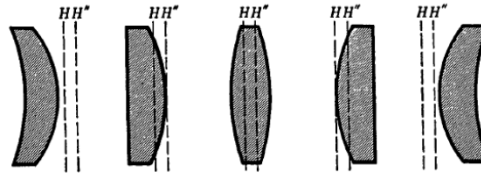


FIGURE 5D
The variation of the positions of the primary and secondary principal planes as a thick lens of fixed focal length is subjected to "bending."

5.4 CONJUGATE RELATIONS

In order to trace any ray through a thick lens, the positions of the focal points and principal points must first be determined. Once this has been done, either graphically or by computation, the parallel-ray construction can be used to locate the image as shown in Fig. 5E. The construction procedure follows that given in Fig. 4M for a thin lens, except that here all rays in the region between the two principal planes are drawn parallel to the axis.

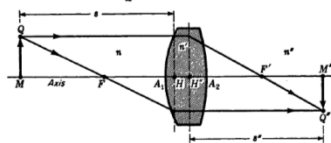


FIGURE 5E
The parallel-ray method of construction for graphically locating an image formed by a thick lens.

By a comparison of the two figures and from the derivations of Eqs. (4n) and (4o), it will be found that, provided the object and image distances are measured to or from the principal points, we can apply the gaussian lens formula

$$\frac{n}{s} + \frac{n''}{s''} = \frac{n}{f} = \frac{n''}{f''} \quad (5c)$$

or by Eq. (3h)

$$V + V'' = P$$

In the special case where the media on the two sides of the lens are the same, so that $n'' = n$, we find $f'' = f$ and Eq. (5c) becomes

$$\frac{1}{s} + \frac{1}{s''} = \frac{1}{f} = \frac{1}{f''} \quad (5d)$$

The first step is to calculate the primary and secondary focal lengths of each surface separately by the use of the formulas for a single spherical surface [Eqs. (3c) and (3d)]. Using the present notation, these are

$$\bullet \quad \frac{n}{f_1} = \frac{n'}{f_1'} = \frac{n' - n}{r_1} \quad \text{and} \quad \frac{n'}{f_2'} = \frac{n''}{f_2''} = \frac{n'' - n'}{r_2} \quad (5f)$$

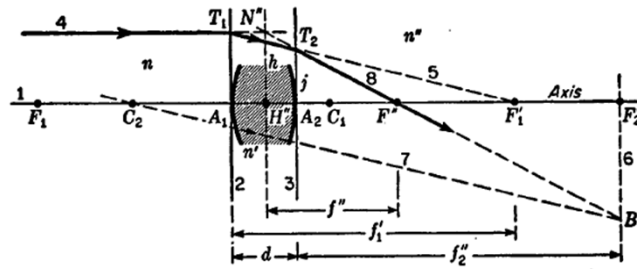


FIGURE 5G
The oblique-ray method for graphically tracing paraxial rays through a thick lens.

With these focal lengths known, the lens axis can be drawn as in Fig. 5G and the known points measured off to some suitable scale. After drawing lines 2 and 3 through the lens vertices, a parallel incident ray 4 is selected. Upon refraction at the first surface the ray takes the new direction 5 toward F'_{11} , the secondary focal point of that surface. After line 6 is drawn through F''_{22} , line 7 is drawn through C_2 parallel to ray 5. The point B where line 6 crosses line 7 determines the direction of the final refracted ray 8. The intersection of ray 8 with the axis locates the secondary focal point F'' of the lens, while its intersection N'' with the incident ray locates the corresponding secondary principal plane H'' .

5.6 GENERAL THICK-LENS FORMULAS

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Gaussian formulas

$$\frac{n}{f} = \frac{n'}{f'_1} + \frac{n''}{f'_2} - \frac{dn''}{f'_1 f'_2} = \frac{n''}{f''}$$

$$A_1 F = -f \left(1 - \frac{d}{f'_2} \right)$$

$$A_1 H = +f \frac{d}{f'_2}$$

$$A_2 F'' = +f'' \left(1 - \frac{d}{f'_1} \right)$$

$$A_2 H'' = -f'' \frac{d}{f'_1}$$

Power formulas

$$P = P_1 + P_2 - \frac{d}{n'} P_1 P_2 \quad (5g)$$

$$A_1 F = -\frac{n}{P} \left(1 - \frac{d}{n'} P_2 \right) \quad (5h)$$

$$A_1 H = +\frac{n}{P} \frac{d}{n'} P_2 \quad (5i)$$

$$A_2 F'' = +\frac{n''}{P} \left(1 - \frac{d}{n'} P_1 \right) \quad (5j)$$

$$A_2 H'' = -\frac{n''}{P} \frac{d}{n'} P_1 \quad (5k)$$

These equations are derived from geometrical relations that can be obtained from a diagram like Fig. 5G.

EXAMPLE 2 A lens has the following specifications: $r_1 = +1.5$ cm, $r_2 = +1.5$ cm, $d = 2.0$ cm, $n = 1.00$, $n' = 1.60$, and $n'' = 1.30$. Find (a) the primary and secondary focal lengths of the separate surfaces, (b) the primary and secondary focal lengths of the system, and (c) the primary and secondary principal points.

SOLUTION (a) To apply the gaussian formulas, we first calculate the individual focal lengths of the surfaces by means of Eq. (5f).

$$\begin{aligned} \frac{n}{f_1} &= \frac{n' - n}{r_1} = \frac{1.60 - 1.00}{1.5} & f_1 &= \frac{1.00}{0.40} = +2.50 \text{ cm} \\ &= 0.400 & f_1' &= \frac{1.60}{0.40} = +4.00 \text{ cm} \\ \frac{n'}{f_2'} &= \frac{n'' - n'}{r_2} = \frac{1.30 - 1.60}{1.5} & f_2' &= \frac{1.60}{-0.20} = -8.00 \text{ cm} \\ &= -0.200 & f_2'' &= \frac{1.30}{-0.20} = -6.50 \text{ cm} \end{aligned}$$

(b) The focal lengths of the system are calculated from Eq. (5g).

$$\begin{aligned} \frac{n}{f} &= \frac{n'}{f_1} + \frac{n''}{f_2'} - \frac{d}{f_1' f_2'} = \frac{1.60}{4.00} + \frac{1.30}{-6.50} - \frac{2.00}{4.00 \cdot -6.50} \\ \frac{n}{f} &= 0.40 - 0.20 + 0.10 = 0.30 \\ \text{or } f &= \frac{1.00}{0.30} = +3.333 \text{ cm} \quad \text{and} \quad f'' = \frac{n''}{0.30} = \frac{1.30}{0.30} = +4.333 \text{ cm} \end{aligned}$$

The focal points of the system are given by Eqs. (5h) and (5j).

$$\begin{aligned} A_1 F &= -f \left(1 - \frac{d}{f_2'} \right) = -3.333 \left(1 - \frac{2.0}{-8.0} \right) = -4.166 \text{ cm} \\ A_2 F'' &= +f'' \left(1 - \frac{d}{f_1'} \right) = +4.33 \left(1 - \frac{2.0}{4.0} \right) = +2.167 \text{ cm} \end{aligned}$$

(c) The principal points are given by Eqs. (5i) and (5k).

$$\begin{aligned} A_1 H &= +f \frac{d}{f_2'} = +3.33 \frac{2.0}{-8.0} = -0.833 \text{ cm} \\ A_2 H'' &= -f'' \frac{d}{f_1'} = -4.33 \frac{2.0}{4.0} = -2.167 \text{ cm} \end{aligned}$$

The graphical solution of this same problem is shown in Fig. 5H.

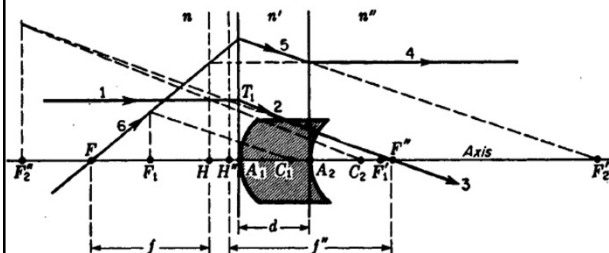


FIGURE 5H
A graphical construction for locating the focal points and principal points of a thick lens.

5.7 SPECIAL THICK LENSES

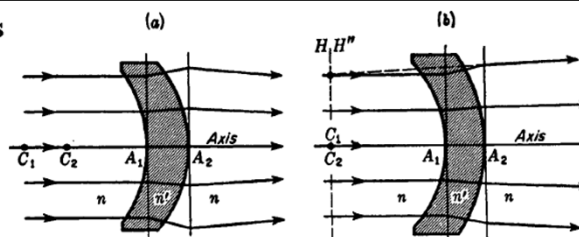


FIGURE 5I
Special thick lenses: (a) a positive lens with equal radii of curvature; (b) a negative lens with concentric surfaces.

Two special lenses of some interest as well as practical importance are presented here.

The first, as shown in Fig. 5I, is a lens with spherical surfaces of equal radii, $r_1 = r_2$. A lens of this description, surrounded by a medium of lower index, $n' > n$, has a small but positive power. Its principal planes are located some distance from and to the right of the lens, and their spacing HH'' is equal to the lens thickness d . If the surrounding medium has a greater index, as in the case of an air space between the surfaces of two lenses of equal index, $n' < n$, the power is again positive but the principal planes lie some distance to the left of the lens and a distance d apart.

The second special case is that of a *concentric lens*, both surfaces having the same center of curvature. Where such a lens is surrounded by a medium of lower index, $n' > n$, the system has a negative power with a long focal length and the principal points coincide with the common center of curvature of the two surfaces. In other words, it acts like a thin lens located at C_1C_2 .

5.8 NODAL POINTS AND OPTICAL CENTER

Of all the rays that pass through a lens from an off-axis object point to its corresponding image point, there will always be one for which the direction of the ray in the image space is the same as that in the object space; i.e., the segments of the ray before reaching the lens, and after leaving it, are parallel. The two points at which these segments, if projected, intersect the axis are called the *nodal points*, and the transverse planes through them are called the *nodal planes*. This third pair of points and their associated planes are shown in Fig. 5J, which also shows the optical center of the lens at C . It is readily shown that if the medium on both sides is the same, the nodal points N and N'' coincide with the principal points H and H'' but if the two media have different indices, the principal points and the nodal points will be separate. Since the incident and emergent rays make equal angles with the axis, the nodal points are called conjugate points of *unit positive angular magnification*.

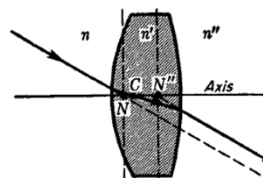


FIGURE 5J
The significance of the nodal points and nodal planes of a thick lens.

If the ray is to emerge parallel to its original direction, the two surface elements of the lens where it enters and leaves must be parallel to each other so that the effect is like that of a plane-parallel plate. A line between these two points crosses the axis at the optical center C .

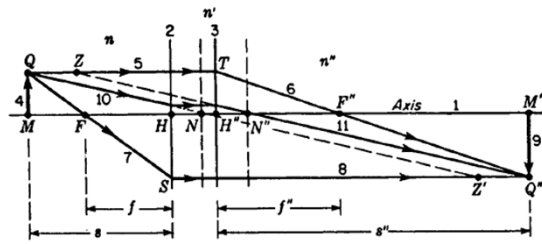


FIGURE 5K

The parallel-ray method of graphically locating the nodal points and planes of a thick lens.

for $n'' \neq n$,

it is observed that the distance between nodal planes is exactly equal to the distance between principal planes. In general, therefore,

$$NN'' = HH'' \quad (5o)$$

the focal lengths, which are measured from the principal points,

$$f = FH = N''F'' \quad \text{and} \quad f'' = H''F'' = FN \quad (5p)$$

the lateral magnification y'/y is given by

$$m = \frac{y'}{y} = -\frac{s'' - HN}{s + HN} \quad (5q)$$

The distance from the first vertex to the primary nodal point

$$HN = f'' \frac{n'' - n}{n''} \quad (5r)$$

$$A_1N = f \left(\frac{d}{f_2} + \frac{n'' - n}{n} \right) \quad (5s)$$

5.9 OTHER CARDINAL POINTS

In thick-lens problems a knowledge of the six cardinal points, comprising the focal points, principal points, and nodal points, is always adequate to obtain solutions.

Other points of lesser importance but still of some interest are (1) negative principal points and (2) negative nodal points. *Negative principal points* are conjugate points for which the lateral magnification is unity and negative. For a lens in air they lie at twice the focal length and on opposite sides of the lens. *Negative nodal points* lie as far from the focal points as the ordinary cardinal nodal points but on opposite sides.

5.10 THIN-LENS COMBINATION AS A THICK LENS

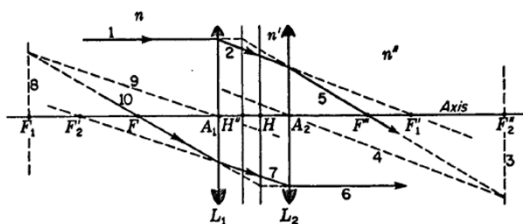


FIGURE 5L
Focal points and principal points of a system involving two thin lenses.

A combination of two or more thin lenses may also be referred to as a thick lens because the optical properties of a set of coaxially mounted lenses can be conveniently treated in terms of only two focal points and two principal points. If the object space and image space have the same refractive index (and this is nearly always the case), the nodal points and planes coincide with the principal points and planes.

The positions of the cardinal points of a combination of two thin lenses in air can be calculated by means of the thick-lens formulas given in Sec. 5.6. As used for thin lenses in place of individual refracting surfaces, A_1 and A_2 become the two lens centers, while f_1 , f_2 and P_1 , P_2 become their separate focal lengths and powers, respectively. The latter are given by

$$P_1 = \frac{n_1 - n}{r_1} + \frac{n' - n_1}{r'_1} = \frac{n}{f_1} \quad P_2 = \frac{n_2 - n'}{r_2} + \frac{n'' - n_2}{r'_2} = \frac{n'}{f'_2} \quad (5t)$$

5.11 THICK-LENS COMBINATIONS

The problem of calculating the positions of the cardinal points of a thick lens consisting of a combination of several component lenses of appreciable thickness is one of considerable difficulty, but one which can be solved by use of the principles already given. In a combination of two lenses such as that in Fig. 5L, if the individual lenses cannot be considered as thin, each must be represented by a pair of principal planes. There are thus two pairs of principal points, H_1 and H'_1 for the first lens and H_2 and H'_2 for the second, and the problem is to combine these to find a single pair H and H' for the combination and to determine the focal lengths. By carrying out a construction similar to Fig. 5G for each lens separately, it is possible to locate the principal points and focal points of each. Then the construction of Fig. 5L can be accomplished, taking account of the unit magnification between principal planes.